

Fig. 7: FASTCUDA block diagram

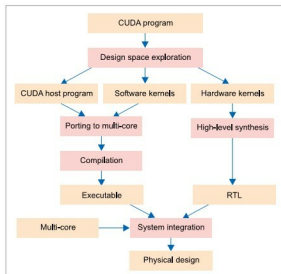


Fig. 8: FASTCUDA toolset

Threads within a thread-block are synchronised, and executed by a single streaming multiprocessor inside a GPU. These share data through a fast and small private memory of the streaming multiprocessor, called 'shared memory'. On the other hand, synchronisation between threads belonging to different thread-blocks is not supported. However, a slow and large 'global memory' is accessible by all thread-blocks.

Similar to a GPU, FASTCUDA employs two separate memory spaces (global and local) as well as a similar mapping of the block-threads onto the FPGA resources.

Bringing software and hardware close together, FASTCUDA accelerates execution of CUDA programs by running some of the kernels in hardware. A state-of-the-art FPGA will provide all the required resources; multiple embedded micro-CPU for the host program and software kernels, and logic capacity for hardware kernels.

Fig. 7 shows the block diagram of the overall FASTCUDA system. A multi-core processor, consisting of multiple embedded cores (configurable small processors), is used to run the host program serially and software kernels in parallel. Threads belonging to the same CUDA thread-block are executed by the same core. Hardware kernels are partitioned into thread-blocks, and synthesised and implemented inside an 'accelerator' block. Each thread-block has a local private memory, while the global shared

memory can be accessed by any thread following the philosophy of the CUDA model.

The FASTCUDA toolset (Fig. 8) is responsible for automating most of this process, thus minimising user intervention.

Design space exploration

The first step is to decide how to make the best use of the available FPGA resources for a given CUDA program. The next is to know what percentage of the FPGA real estate should be allocated to the multi-core processor for software kernels, and what percentage should be allocated to the accelerator for hardware

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